PROPER FUNCTIONAL REPRESENTATION OF SYMMETRIC DOUBLE MINIMUM POTENTIALS

Eugene M. FLUDER Jr. and José R. DE LA VEGA
Department of Chemistry, Villanova University, Villanova, Pennsylvania 19085, USA

Received 18 August 1978

The suitability of using either a parabola and gaussian or two-Morse potentials as the analytic function for a symmetric double-minimum potential is examined. It is determined that the choice must be made by evaluating a simple expression. An expression for the matrix elements of the secular equation using the two-Morse potential is given.

1. Introduction

The eigenstates of a double-minimum potential have recently been used to calculate tunnelling in proton transfer reactions [1–4]. The eigenfunctions for the double-minimum potential were expanded in terms of the harmonic oscillator eigenfunctions. The variational method was used to calculate the eigenvalues and eigenvectors which were then used to calculate the time evolution of the system.

For computational convenience, a fourth degree polynomial [5] was used to describe such a potential in order to determine the eigenstates. It has been found [2–4,6,7] that the double-minimum potentials obtained from ab initio calculations for various proton transfers frequently were not well represented by a fourth degree polynomial. In these situations, a potential function which is the sum of a parabolic and a gaussian function was used.

Recent investigations [3], however, have resulted in symmetric double-minimum potentials that could not be fitted to a parabola and gaussian potential either. It was determined that, in these cases, an analytic function which is the sum of two Morse potentials [8] should be used.

The present analysis of the suitability of the parabola and gaussian or two-Morse potential as a fitting function for a symmetric double-minimum potential shows that the choice is not arbitrary. The features of the potential determine, by means of a simple inequality, which of the two functions must be used to the exclusion of the other.

2. Parabola and gaussian

A symmetric double-minimum potential may be represented as the sum of a parabola and a gaussian function:

$$V(x) = a_2x^2 + v_0 \exp(-\alpha x^2). \tag{1}$$

Three parameters are used to fit this function to a given symmetric double-minimum potential: a_2, v_0, and α. These parameters are uniquely determined by the energy barrier, E, the interminimal distance, D, and the curvature at the minima, k_m. The equations relating these features to the potential are:

$$a_2 = \alpha v_0 \exp(-\alpha x_m^2), \tag{2}$$

$$k_m = 4a_2\alpha x_m^2, \tag{3}$$

$$E = (a_2/\alpha)\,[\exp(\alpha x_m^2) - 1 - \alpha x_m^2]\,, \tag{4}$$

where $x_m = \pm D/2$.

By solving eq. (3) for a_2 and substituting this value into eq. (4), the transcendental equation for α is obtained:

$$\alpha = (k_m/ED^2)^{1/2}\,[\exp(\alpha x_m^2) - 1 - \alpha x_m^2]^{1/2}. \tag{5}$$

Although eq. (5) uniquely determines the value of α, this value will be real if, and only if,

$$\tfrac{1}{4}D\,(k_m/2E)^{1/2} < 1. \tag{6}$$

3. Two Morse potentials

In the situations where condition (6) cannot be met, the sum of two Morse potentials must be used. In the symmetric case, the function can be written:

$$V(\xi) = v_0 - [2v_0 \exp(-\xi_0)] \cosh(\xi)$$
$$+ [v_0 \exp(-2\xi_0)] \cosh(2\xi),$$

where $\xi = \beta x$ and $\xi_0 = \beta x_0$.

The parameters v_0, β, and x_0, which define the function can again be uniquely determined from E, D, and k_m. The equations used are:

$$\cosh(\xi_m) = \tfrac{1}{2} \exp(\xi_0), \tag{7}$$

$$\xi_m = \pm \beta D/2,$$

$$E = \tfrac{1}{2} v_0 [1 - 2 \exp(-\xi_0)]^2, \tag{8}$$

$$k_m = \beta^2 v_0 [1 - 4 \exp(-2\xi_0)]. \tag{9}$$

From these, the transcendental equation for β results:

$$\beta = \left(\frac{k_m}{2E}\right)^{1/2} \left(\frac{1 - 2 \exp(-\xi_0)}{1 + 2 \exp(-\xi_0^0)}\right)^{1/2}. \tag{10}$$

This equation admits of a real solution if, and only if,

$$\tfrac{1}{4} D (k_m/2E)^{1/2} > 1. \tag{11}$$

4. Discussion

It can be seen, by comparing conditions (6) and (11), that the parabola and gaussian and the two-Morse potential are mutually exclusive as analytic functions which represent a symmetric double-minimum potential. The choice is determined by the values of the energy barrier, interminimal distance and the curvature at the minima. It can be seen from fig. 1 that, in general, the parabola and gaussian can be used when the curvature at the minima is small while the two-Morse potential will be a satisfactory representation when the curvature is large.

The solutions to the resulting Schrödinger equation can in each case be expanded as linear combinations of harmonic oscillator eigenfunctions. The matrix elements for the parabola and gaussian potential have been reported previously [3]. The matrix elements for the two-Morse potential are:

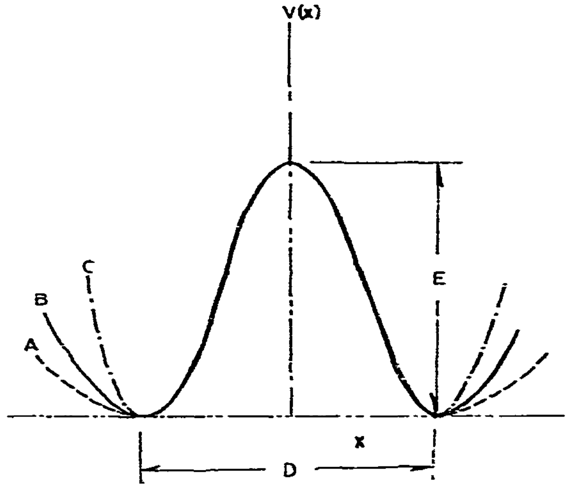

Fig. 1. Comparison of a parabola and gaussian potential (A) and a two-Morse potential (C) with a fourth-degree polynomial (B). For a given energy barrier and interminimal distance, a parabola and gaussian potential cannot be used when $k_m > 32\,E/D^2$ and a two-Morse potential cannot be used when $k_m < 32\,E/D^2$.

$$H_{n,m} = (n + \tfrac{1}{2})\gamma\delta_{n,m} - \tfrac{1}{2}\gamma[n(n-1)]^{1/2}\delta_{m,n-2}$$
$$- \tfrac{1}{2}\gamma[(n+1)(n+2)]^{1/2}\delta_{m,n+2}$$
$$+ N_n N_m \sum_{k=\substack{0\\1}}^{n} \sum_{l=\substack{0\\1}}^{m} \{[a(n,k)a(m,l)/2^{k+l}]$$
$$\times [B \exp(-1/\gamma) H^*_{k+l}(\gamma^{-1/2})$$
$$- A \exp(-1/4\gamma) H^*_{k+l}((4\gamma)^{-1/2})]\},$$

$$N_n = (2^n n!)^{-1/2}, \quad A = 2mv_0 \exp(-\xi_0)/\beta^2\hbar^2,$$

$$B = mv_0 \exp(-2\xi_0)/\beta^2\hbar^2,$$

where the summations over k and l run by increments of 2 and start at 0 or 1, depending on whether the upper limit is even or odd, and where γ is a variational parameter, $a(i,j)$ is the jth coefficient of the Hermite polynomial of order i and $H_i^*(x)$ is the Hermite polynomial of order i with all coefficients positive evaluated at x.

Acknowledgement

The authors wish to thank Jan H. Busch for his valuable discussions.

References

[1] J. Brickmann and H. Zimmerman, J. Chem. Phys. 50 (1969) 1608.
[2] J.H. Busch and J.R. de la Vega, J. Am. Chem. Soc. 99 (1977) 2397.
[3] E.M. Fluder and J.R. de la Vega, J. Am. Chem. Soc. 100 (1978) 5265.
[4] J.D. Swalen and J.A. Ibers, J. Chem. Phys. 36 (1962) 1914.
[5] R.L. Somorjai and D.F. Hornig, J. Chem. Phys. 36 (1962) 1980.
[6] S.I. Chan, J. Zinn, J. Fernandez and W. Gwinn, J. Chem. Phys. 33 (1960) 1643.
[7] M.C. Flanigan and J.R. de la Vega, J. Chem. Phys. 61 (1974) 1882.
[8] P.-O. Löwdin, Advan. Quantum Chem. 2 (1965) 254.